

STATISTICS

is not the answer.

It is a way to ask better questions of your data.

BIC8025 | Introduction to Statistical Analysis

NTNU • Fall 2026

Is this course for you?

Introduction to *statistical analyses*, not *statistics*

Feature	Introduction to statistics	Introduction to statistical analysis
Focus	Basic concepts, probability theory, and classic statistical models	Applying statistical concepts and models to real data, and draw scientific conclusions
Topics	Descriptive statistics (mean, variance, SD, SE); sampling distribution; t-test; ANOVA; simple correlation; simple regression	Data exploration; whatever models, simple or complicated, that are appropriate for the data; model diagnosis
Punchline	What is a car (What is statistics)?	How to drive a car (How to use statistics to get to where I want)?

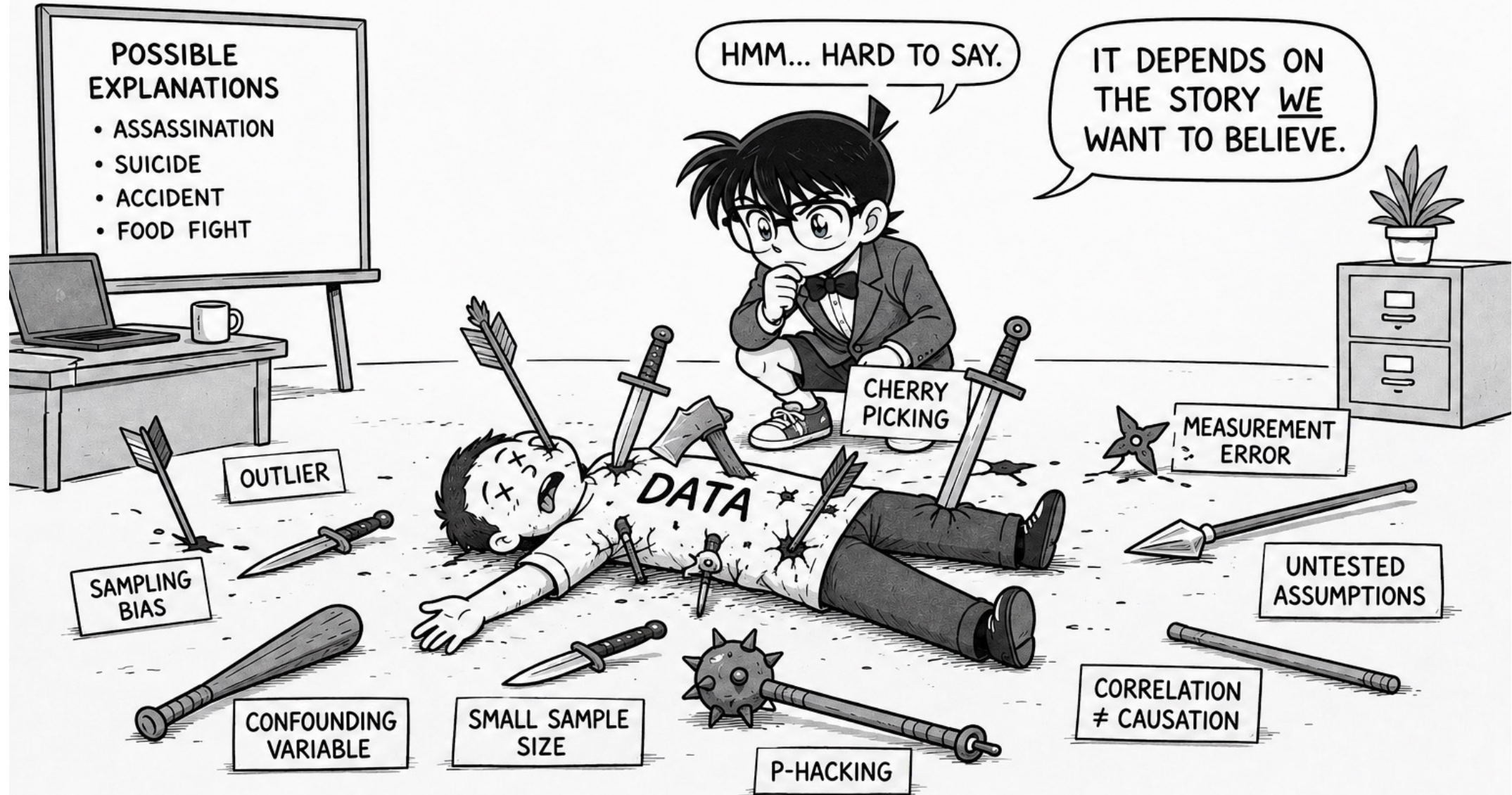
What this course is about

Moving from statistical recipes to scientific reasoning

- Start with a scientific question.
- Think about how observations were generated.
- Choose models that match the structure of the data.
- Use statistics to **investigate, not just calculate**.
- Build analyses that can support publication-quality research.

DATA DO NOT SPEAK FOR THEMSELVES.

WE DECIDE WHAT STORY THEY TELL.

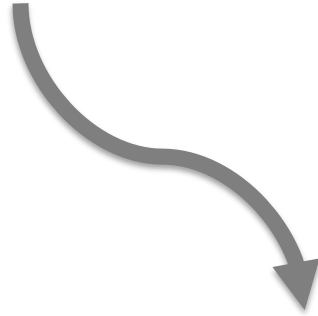


By the end of the semester, you should be able to...

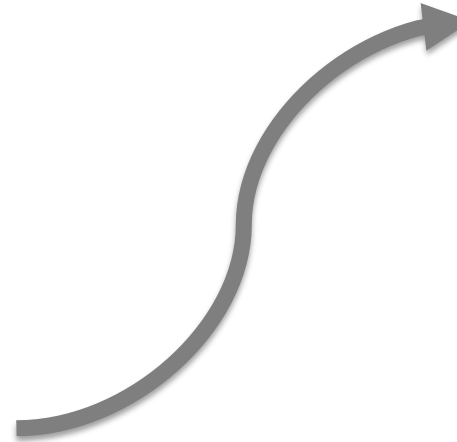
- Design experiments and analyses that address the questions.
- Recognize the assumptions and limitations behind statistical models.
- Use R and statistical software to analyze real data.
- Interpret model results in biological—not merely statistical—terms.
- **Develop and defend** an analysis of your own research data.

The central idea

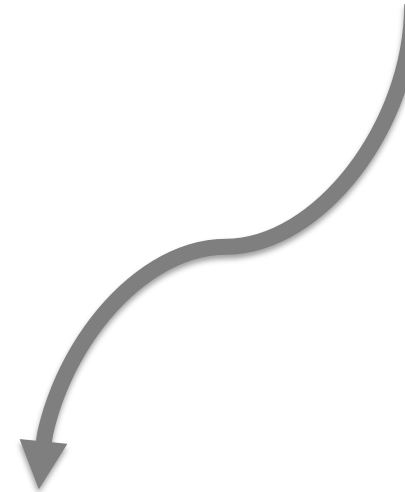
Biological process
that generated data



Sampling process
that generated your data



Statistical model
that is a simplified version
of biological process



Scientific inference
that you believe



WEEK 1-4:
SAMPLING &
RESAMPLING



WEEK 5:
MINI-PROJECT I



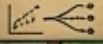
WEEK 6-9:
LINEAR MODELS



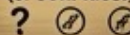
WEEK 10:
MINI-PROJECT H



WEEK 11-13:
MIXED MODELS



WEEK 14:
MODEL SELECTION
(or SURPRISE!)



WEEK 15-16:
FINAL PROJECT
PRESENTATION



Mei's Statistics
Course Journal:
16-Week:
統計學
STATISTICS
WEEK 0:
HIKING BEGINS!



WEEK 5:
MINI-PROJECT I

WEEK 1-4:
SAMPLING &
RESAMPLING

WEEK 5:
MINI-PROJECT I

WEEK 6-9:
LINEAR MODELS

WEEK 10:
MINI-PROJECT II

WEEK 11-13:
MIXED MODELS

WEEK 14:
MODEL SELECTION
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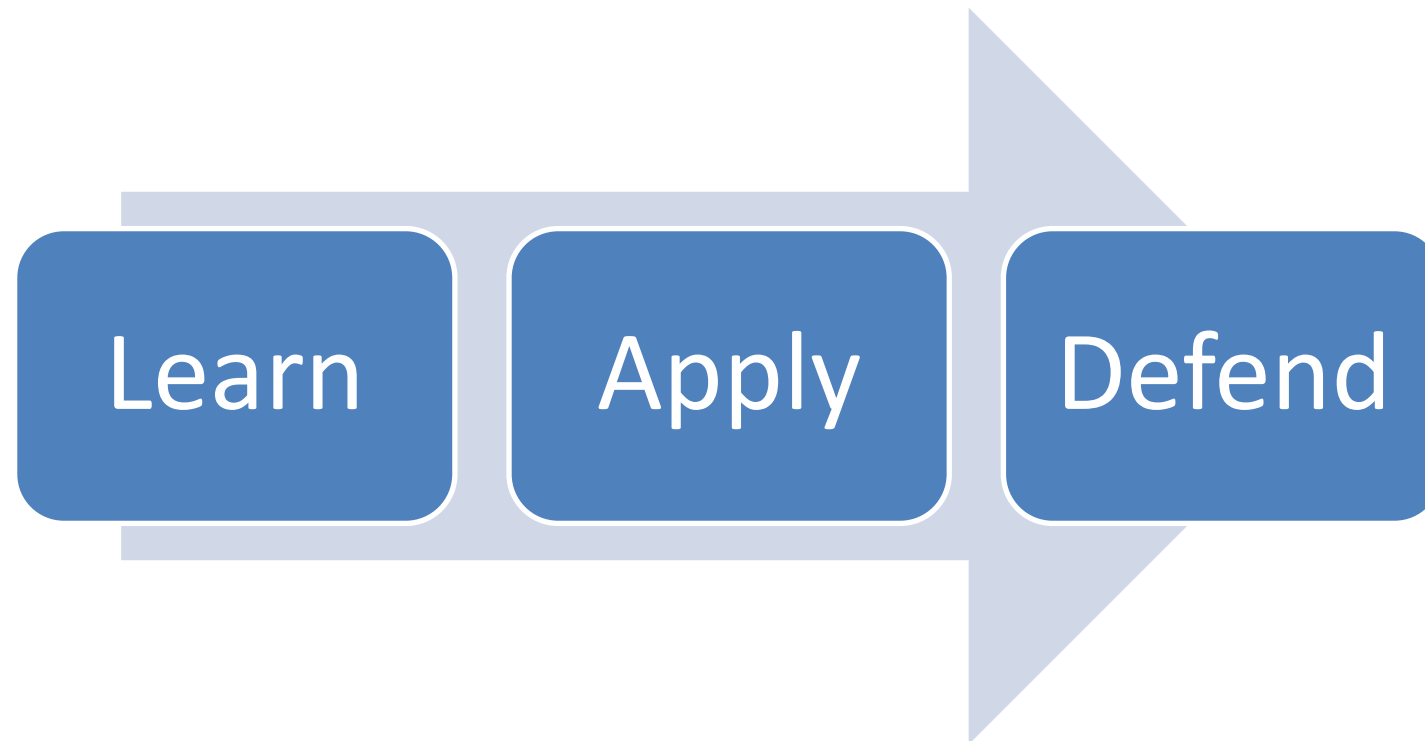
WEEK 15-16:
FINAL I PROJECT
PRESENTATION

Mei's Statistics
Course Journal:
16-Week:
**CRISIS
CHECKLIST.**
WEEK 0:
HERO TRAIL BEGINS!

This is a project-driven course

Two mini-projects + a final project

Ideally, use your own research data for the projects



What will we actually do in class?

- Short lectures to build the conceptual framework.
- Real-world biological examples and case studies.
- Instructor-led discussion and problem-based learning.
- In-class exercises and homework using statistical software.
- **AI-facilitated coding** where it helps—but understanding remains **your responsibility**.

With great power, comes great responsibility

Non-free comic image © Marvel Comics

AI-FACILITATED CODING: GREAT POWER, GREAT RESPONSIBILITY

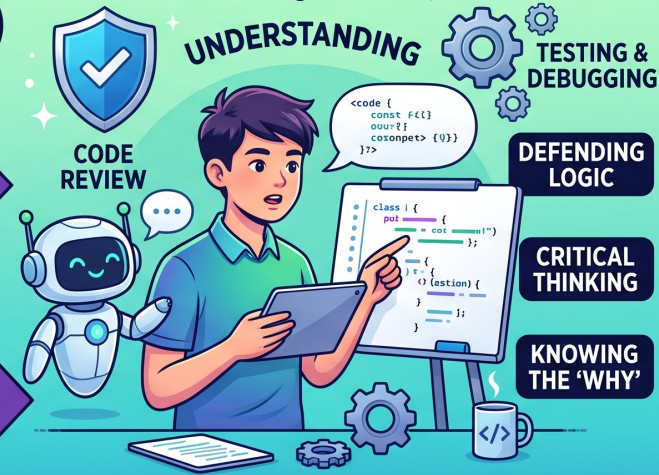
WITH GREAT POWER...

(means AI-generated code and assistance)



...COMES GREAT RESPONSIBILITY

(means YOU are responsible for understanding and defending YOUR code)



BE THE MASTER OF YOUR CODE. APPRECIATE THE AI TOOL, CHERISH THE LEARNING PROCESS.

#AIEducation #ResponsibleCoding



What I expect from you

- Come prepared to think and act (sleep well the night before; learn what mean, variance, t-test etc. on your own time).
- Ask **why** a method is appropriate before asking **how** to run it.
- Bring your own research questions and data whenever possible.
- Be willing to discover that your first model may be wrong.
- Explain and defend your analytical decisions in both statistical and biological language.

Let's begin!

